11. DNS SERVER CONFIGURATION

Course: Diploma in Cyber Security

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Introduction

DNS maps domain names to IP addresses. A DNS server makes it easier for users to open a name such as www.musicmania.in instead of remembering an IP address.

Task 1: Create DNS zone

In a lab/VM, install BIND on Linux or use an appropriate Windows DNS tool and create a zone for musicmania.in. Use a safe isolated lab environment.

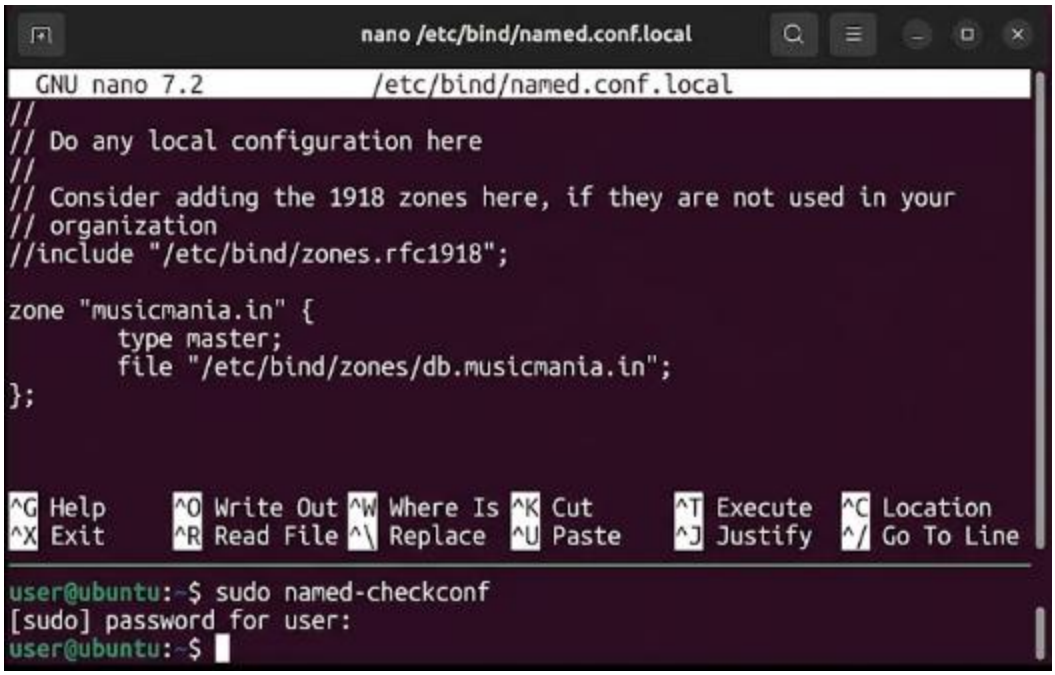
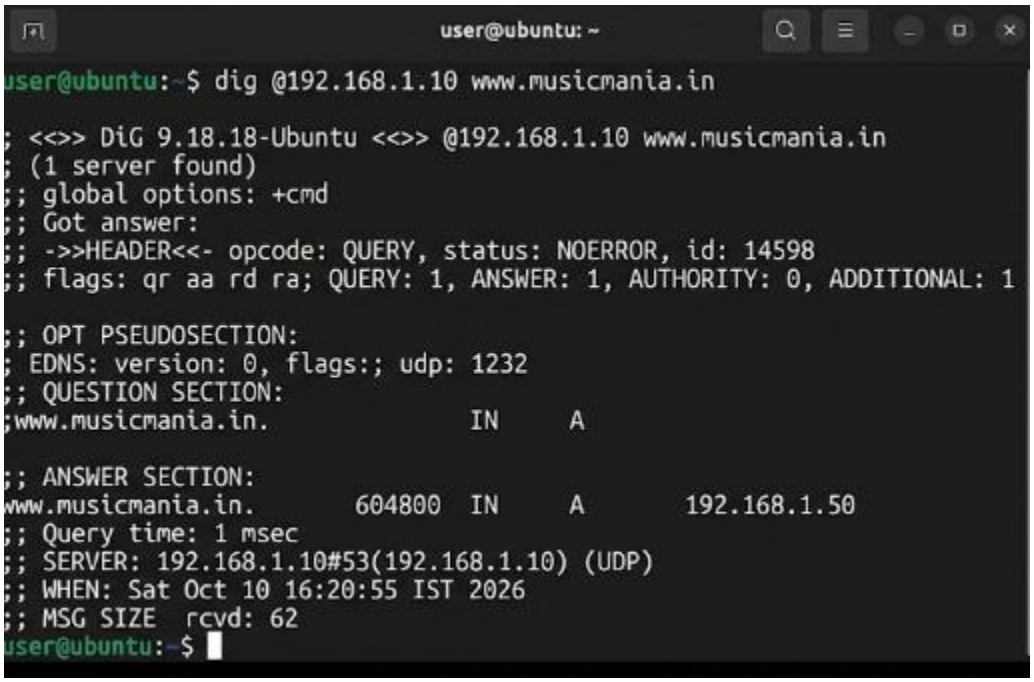


Figure 1: BIND9 Forward Lookup Zone Definition for musicmania.in on Linux Server

Task 2: Add A record

Create an A record mapping www.musicmania.in to 192.168.1.50, then test using nslookup or dig.



Check #	Diagnostic Target	Command Executed	Purpose & Validation Crite

Figure 2: Host (A) Record Resolution Verifying Mapping to Target IP 192.168.1.50

Task 3: CNAME record

Create play.musicmania.in as a CNAME pointing to www.musicmania.in and test that it resolves to the same target address.

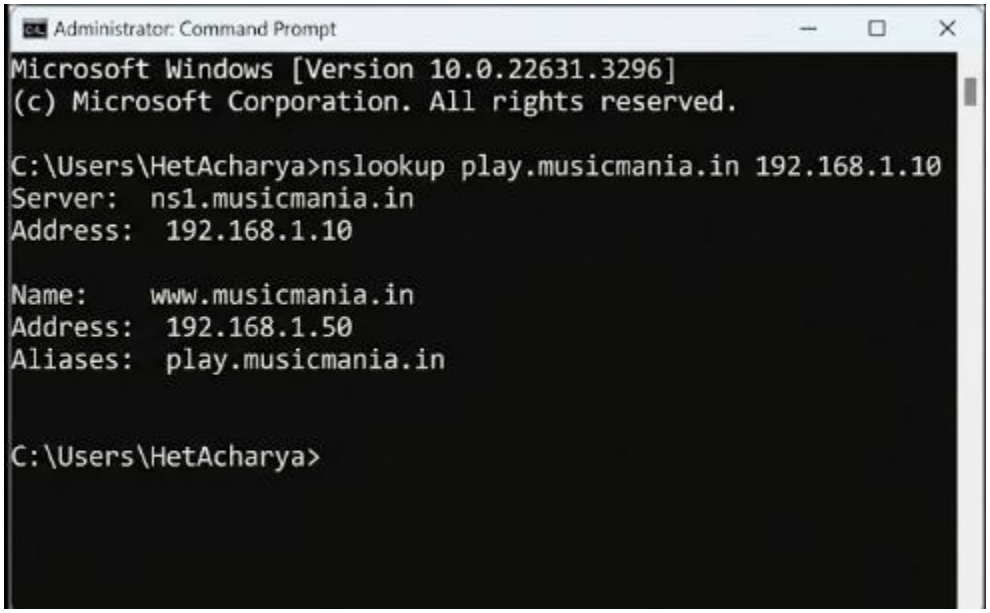


Figure 3: Canonical Name (CNAME) Resolution Chaining play.musicmania.in to 192.168.1.50

Task 4: Troubleshoot DNS

Check the zone/configuration files, service status, restart the DNS service if required, and test again with nslookup or dig. Also c

Check #	Diagnostic Target	Command Executed	Purpose & Validation Crite

Check 1	**Syntax & Zone File Integrity**	`named-checkconf`
		
		`named-checkzone musicmania.in /etc/bind/zones/db.musicmania.in`
			Checks for missing semicolon trailing dots, or formatting erro Output must return `OK` and dis the serial number.

Check 2	**DNS Daemon Operational Status**	`sudo systemctl status bind9` (or `named`)	Verifies that the service state `active (running)` and listening UDP/TCP port 53 without cras faults.
Check 3	**Service Reload / Restart**	`sudo systemctl restart bind9`	Flushes internal cache and re modified zone files after upda serial numbers.
Check 4	**Client Resolver Configuration**	`cat /etc/resolv.conf` (or `ipconfig /all` on Windows)	Confirms the client's primary I pointer is set to `nameserver 192.168.1.10` rather than an ex public DNS (like 8.8.8.8).

heck that the client is using the correct DNS server.

Conclusion

This topic helped me understand the basic concept and how it can be used in a real network. I also understood that practical results should be checked on the actual system or simulator instead of assuming the result.